



High-Impedance AUDIO BUFFER WITH JFET

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The buffer circuit described here can be used to enhance the AC input impedance of audio amplifiers that are used with pickups in musical instruments.

Some signal sources for pre-amplifiers, passive pickups for electric guitars or sensors based on capacitors require very high-impedance of over 5-mega-ohm. This can be achieved easily with junction field effect transistor (JFET) but may require special design of the printed circuit board (PCB), appropriate construction techniques of the box, proper cables and connectors.

This circuit provides a solution for a high-impedance, low-cost, low-quiescent current, buffer-follower based on PN4393 JFET.

Circuit and working

The circuit of the high-impedance audio buffer with JFET is shown in Fig. 1. It has almost unity gain and relatively-low-output impedance. The buffer can be used with any appropriate JFET including J201, J202, J113, PN4391, PN4392, PN4393, 2N5457, 2N5458, 2N3819, BF245B, MPF102 or similar. Although there are some differences between these JFETs, all of these can do the job depending on the input signal range. However, care must be taken because some JFETs are not pin-to-pin compatible.

Audio input is applied to connector CON1. Input impedance depends mainly on the values of resistors R2 and R8 and is around 5-mega-ohm. If resistors R8 and R9 and capacitor C6 are omitted, input resistance will de-

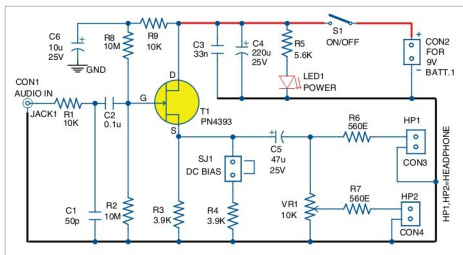


Fig. 1: Circuit diagram of the high-impedance audio buffer with JFET

pend mainly on R2 and will be around 10-mega-ohm.

Value of R2 can be increased but it may be difficult to maintain the high input resistance during the practical usage of the circuit. Resistor R1 and capacitor C1 are used for protection and filtering functions for the input of the buffer, respectively. C1 can be omitted but not R1. Values of R1 and C1 can be changed as per requirement. Capacitor C2 is used to remove DC input components.

Resistors R8 and R9 between gate G and drain D of transistor T1 are not always needed. So resistor R8 is not necessarily equal to R2. Also, R8 will reduce the input impedance. Use of high-value resistors R2 and R8 may increase input noise, so care must be taken while selecting their values.

Resistor R9 and capacitor C6 are for filtering noise from the power source. These can be omitted if the power source is clean. But filtering capacitors C3 and C4 should not be omitted.

PARTS LIST

Semiconductors:

- T1 - PN4393, n-channel JFET
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R9 - 10-kilo-ohm
- R2, R8 - 10-mega-ohm
- R3, R4 - 3.9-kilo-ohm
- R5 - 5.6-kilo-ohm
- R6, R7 - 560-ohm
- VR1 - 10-kilo-ohm potentiometer

Capacitors:

- C1 - 50pF ceramic
- C2 - 0.1 μ F ceramic
- C3 - 33nF ceramic
- C4 - 220 μ F, 25V electrolytic
- C5 - 47 μ F, 25V electrolytic
- C6 - 10 μ F, 25V electrolytic

Miscellaneous:

- S1 - On/off switch
- BATT.1 - 9V, 6F22 or PP3 battery
- CON1 - Audio jack connector
- CON2-CON4 - 2-pin connector
- SJ1 - Shorting jumper
- HP1, HP2 - Headphone

Resistors R3 and R4 provide DC bias for the JFET. Resistor R4 is connected to transistor T1 through shorting jumper SJ1. Range of R3 is from 1-kilo-ohm to 10-kilo-ohm. Lower values will provide lower output impedance but will increase the